

# Ultrafiltration

## Group 2

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Chemical Engineering Lab II

4/9/2019

# Introduction

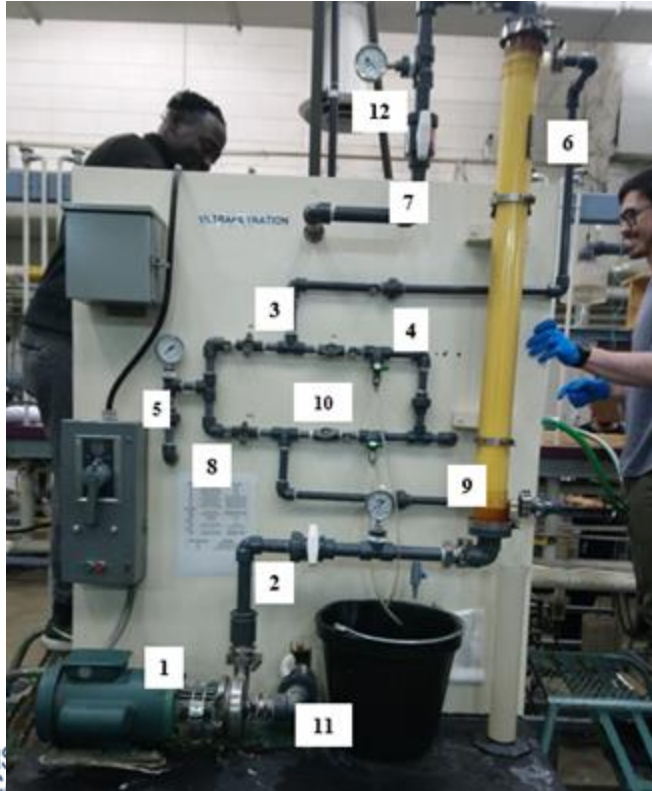
- The purpose of this experiment is to see how ultrafiltration can separate a concentrate of milk from water
- Module includes a feed tank, a permeate milk tank, and a hollow membrane column that consist of fibers outlining the inner diameter of the wall
- Flux of the membrane will be determined by pressure drop across the membrane and from milk concentrations

# Theory

- Solution is pumped across a membrane that is permeable to the solvent but does not allow solutes of a molecular weight above the molecular weight cutoff pass through the membrane
- Separation occurs based on size exclusion or particle capture.
- The optimal operating pressure of the system is determined by graphing the Concentration Factor (CF) versus the flux

$$CF = V_{\text{original}} / V_{\text{concentrate}} = V_{\text{original}} / (V_{\text{original}} - V_{\text{permeate}})$$

# Apparatus



|    | Apparatus | Description                           |
|----|-----------|---------------------------------------|
| 1  | Pump      | Used to Transfer milk solution        |
| 2  | V1        | Inlet Pressure Valve                  |
| 3  | BV3       | Backflush Valve                       |
| 4  | V3        | Permeate Flow Valve                   |
| 5  | P3        | Backflush Pressure Gauge              |
| 6  | Membrane  | Used to filter proteins from solution |
| 7  | V2        | Outlet Pressure Control Valve         |
| 8  | BV4       | Backflush Valve                       |
| 9  | P1        | Inlet Pressure Gauge                  |
| 10 | V4        | Permeate Flow Valve                   |
| 11 | PV1       | Intake-Pump Valve                     |
| 12 | P2        | Outlet Pressure Gauge                 |

# Apparatus



|    | Apparatus     | Description                    |
|----|---------------|--------------------------------|
| 13 | Permeate Tank | Used to Collect Permeate       |
| 14 | V8            | Process Tank Return            |
| 15 | V5            | Permeate Drain                 |
| 16 | V7            | Permeate Drain                 |
| 17 | V6            | Permeate Recycle Valve         |
| 18 | Solution Tank | Holds Solution to be separated |
| 19 | BV1           | Recycle Valve                  |
| 20 | V9            | Process Return Bypass          |
| 21 | BV2           | System Return Valve            |
| 22 | Pump          | Moves Solution through system  |

# Material & Supplies

|    | Materials        | Description                               |
|----|------------------|-------------------------------------------|
| 1  | Water            | Water used in the system                  |
| 2  | Phosphoric Acid  | Acid used in cleaning process             |
| 3  | Sodium Hydroxide | Caustic solution used in cleaning process |
| 4  | Bleach           | Used in Cleaning Process                  |
| 5  | Stirring Rod     | Used to mix fluids                        |
| 6  | Thermometer      | Used to measure temperature               |
| 7  | Litmus Paper     | Used to measure the approximate pH        |
| 8  | pH Meter         | Used to measure the pH more accurately    |
| 9  | Bucket           | Fluid containment equipment               |
| 10 | Milk             | Fluid that will be separated              |
| 11 | Pipet            | Used to transfer solutions                |
| 12 | Beaker           | Fluid containment equipment               |

# Cleaning Time!!

- Morning group performs steps 1-3 in lab manual
- Afternoon group performs steps 3-5 in lab manual
- Morning group performs step 6 in lab manual on day of lab

# Procedure: Water Flux Test

- Add 20L of 104°F deionized water to the tank
- Set inlet pressure to 25 psi
- Set outlet pressure to 5 psi
- Collect permeate for 30 seconds and calculate flux
- Repeat for outlet pressures of 10, 15, and 20 psi

# Procedure: Milk Test

- Add 1 gal of 1% Lowfat milk & 104°F deionized water to the 80L mark of the process tank
- Set inlet pressure to 25 psi
- Set outlet pressure to 5 psi
- Recover 10% (8L) of water to the permeate tank
- Measure the flux for outlet pressures of 5, 10, 15, 20, and 5 psi
- Repeat for recoveries of 25% and 40%

# Experimental Challenges

- Pressure gauge not steady
- Monitor both gauges
- Recovery of 25% and 40% takes a long time
- Need a ladder or tall person to work the valve

# Safety

- Clean up any water spills to prevent slipping hazards
- Pump is mechanical
- Be careful when using step ladders to reach process tank and pressure gauges